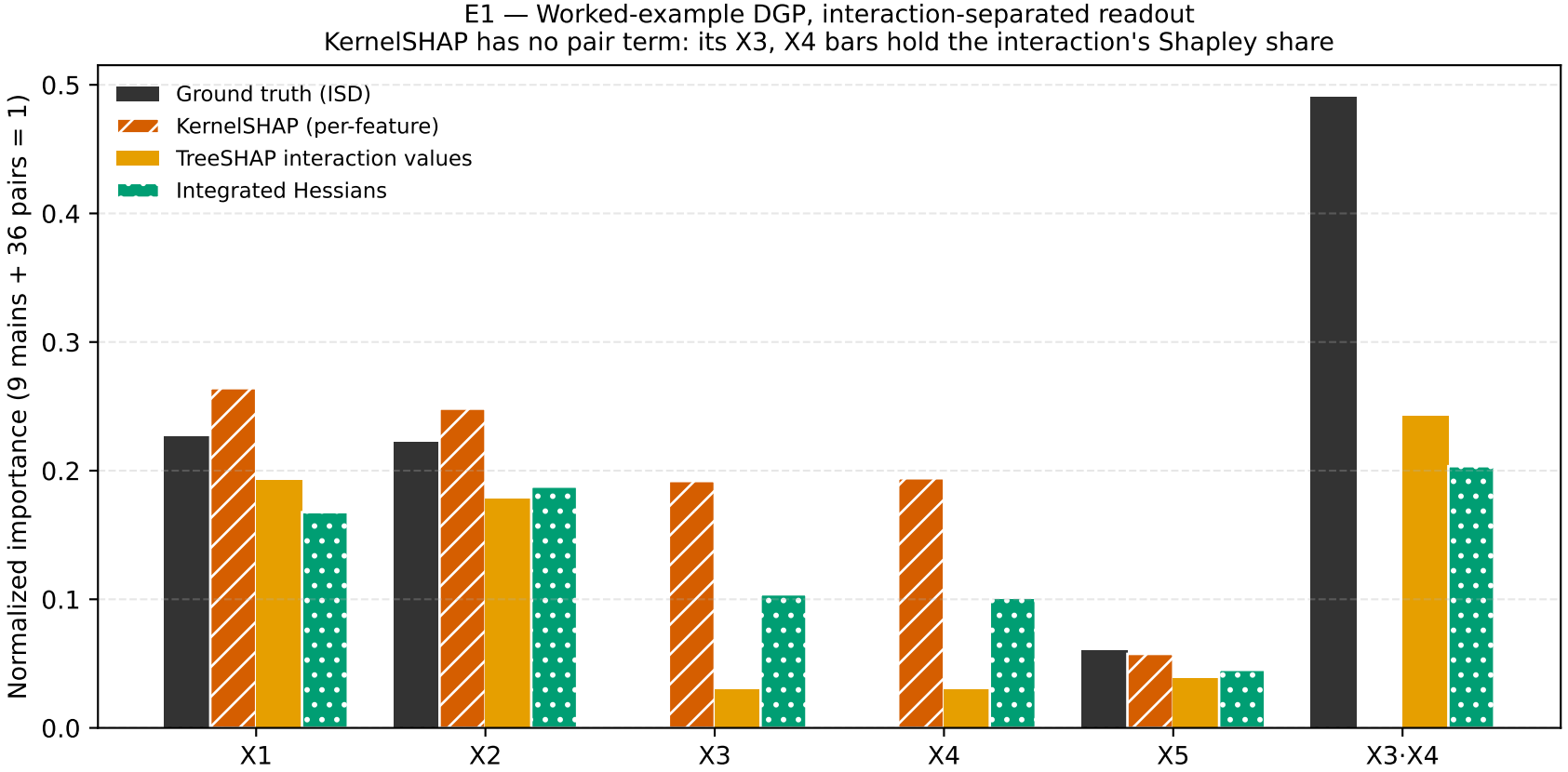
E1 — Worked-example DGP, interaction-separated readout
KernelSHAP has no pair term: its X3, X4 bars hold the interaction's Shapley share
Ground truth (ISD)
KernelSHAP (per-feature)
TreeSHAP interaction values
Integrated Hessians
Normalized importance (9 mains + 36 pairs = 1)
0.0
0.1
0.2
0.3
0.4
0.5
X1
X2
X3
X4
X5
X3·X4